

Colette LaMonica Kelly

(they/them) | Associate Research Scientist

Lamont-Doherty Earth Observatory, Columbia University

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Education

Stanford University

June 2023

Ph.D. in Earth System Science

Dissertation: *Quantifying Pacific Ocean Nitrous Oxide Cycling using Intramolecular Isotope Measurements and Modeling.*

Barnard College of Columbia University

May 2017

B.A. in Environmental Science and Dance (double major)

Phi Beta Kappa, Summa Cum Laude

Theses: *Drivers of Seasonal and Interannual Variability of Waquoit Bay Carbonate Chemistry;*
Dancing Up the Glass Escalator: Institutional Advantages for Men In Ballet Choreography.

Professional Experience

Associated Research Scientist

2026-present

Lamont-Doherty Earth Observatory, Columbia Climate School, Palisades, NY

U.S. GO-SHIP Postdoc Fellow

2024-2026

Woods Hole Oceanographic Institution, Woods Hole, MA

Supervisors: David Nicholson & Alison Macdonald

WHOI Postdoc Scholar

2023-2024

Woods Hole Oceanographic Institution, Woods Hole, MA

Supervisors: David Nicholson & Alison Macdonald

Graduate Research Assistant

2017-2023

Stanford University, Stanford, CA

Supervisor: Karen L. Casciotti

Summer Student Fellow

2016

Woods Hole Oceanographic Institution, Woods Hole, MA

Supervisor: Daniel McCorkle

Undergraduate Research Assistant

2014-2016

Barnard College, New York, NY and Lamont-Doherty Earth Observatory, Palisades, NY

Supervisor: Brian Mailloux

Research Cruises

GO-SHIP P16C

upcoming

R/V Thomas G. Thompson, University of Washington

Papeete, Tahiti to Honolulu, Hawai'i

Contributions: Chief scientist

GO-SHIP A16S

3/9/2026-5/1/2026

R/V Roger Revelle, Scripps Institution of Oceanography

Punta Arenas, Chile to Mindelo, Cabo Verde

Chief Scientist: Leticia Barbero

Contributions: Co-chief scientist

Nitrogen cycling in the eastern tropical South Pacific

11/18/2023-12/20/2023

R/V Roger Revelle, Scripps Institution of Oceanography

Valparaiso, Chile to Arica, Chile

Chief Scientist: Bess B. Ward

Contributions: deployment of core and BGC Argo floats, collection of BGC-Argo float validation samples; collection of samples for N_2O isotopomer analysis and NO_3^- , and NO_2^- isotopic analysis; running incubations conducted with ^{15}N tracer to determine N_2O production rates.

GEOTRACES Section GP15

9/18/18-11/25/18

R/V Roger Revelle, Scripps Institution of Oceanography

Seattle, WA to Papeete, Tahiti

Chief Scientists: Karen Casciotti, Phoebe Lam, and Greg Cutter

Contributions: ODF "supertech": collection of samples for long-lived thorium isotopes and protactinium, neodymium and rare earth elements, silicon isotopes, and stable oxygen isotopes of water; collection of samples for N_2O isotopomer analysis and NO_3^- , and NO_2^- isotopic analysis.

Measuring rates with in-situ incubators

6/28/2018-7/21/2018

R/V Falkor, Schmidt Ocean Institute

San Diego, CA to San Diego, CA

Chief Scientists: Karen Casciotti and Andrew Babbin

Contributions: operation of in-situ incubator for nitrogen transformation experiments; collection of samples for N_2O isotopomer analysis and NO_3^- , and NO_2^- isotopic analysis; running incubations conducted with ^{15}N tracer to determine N_2O production rates; assisting in onboard $[NO_2^-]$ analyses via spectrophotometer.

Nitrogen cycling in the eastern tropical North Pacific

3/16/2018-4/17/2018

R/V Sally Ride, Scripps Institution of Oceanography

San Diego, CA to Manzanillo, Mexico

Chief Scientist: Bess B. Ward

Contributions: collection of samples for N_2O isotopomer analysis and NO_3^- , and NO_2^- isotopic analysis; running incubations conducted with ^{15}N tracer to determine N_2O production rates; assisting in onboard $[NO_2^-]$ analyses via spectrophotometer.

Publications

(* = Authored by undergraduate mentee)

17. C.L. Kelly, B.X. Chang, J. Bullister, M. Warner, N. Yoshida S. Toyoda, A. Bourbonnais, M. de la Paz Arandiga, A. Makabe, S. Kawagucci, A. Barna, A.M. Macdonald¹, & D.P. Nicholson (*in prep*). A Global Ocean Nitrous Oxide Atlas from the GO-SHIP Program.
16. M. Bif, C.L. Kelly, M. Altabet, A. Bourbonnais, C. Elbon, E. Flores, K. Johnson, A. Mnich, J. Plant & K.S. Johnson (2026). BGC-Argo float reveals shifts in nitrogen-carbon cycling in an oxygen-deficient zone. *Communications Earth & Environment*, 7(1). <https://doi.org/10.1038/s43247-026-03410-5>
15. C.L. Kelly, B.X. Chang, A. Emmanuelli, E. Park, A. Macdonald, & D.P. Nicholson (2026). Low-pressure storms drive nitrous oxide emissions in the Southern Ocean. *Nature Communications*, 17(1). <https://doi.org/10.1038/s41467-026-68744-2>
14. X. Sun, C. Frey, D. McCoy, M.B.A. Spieler, C.L. Kelly, A.E. Maloney, E. Garcia-Robledo, M.F. Lehmann, B.B. Ward, & E.J. Zakem (2025). Mechanistic understanding of nitrate reduction as the dominant production pathway of nitrous oxide in marine oxygen minimum zones. *Nature Communications*, 16(1). <https://doi.org/10.1038/s41467-025-63989-9>
13. C.L. Kelly, N.M. Travis, P.A. Baya, C. Frey, X. Sun, B. B. Ward, & K. L. Casciotti (2024). Isotopomer labeling and oxygen dependence of hybrid nitrous oxide production. *Biogeosciences*, 21(14). <https://doi.org/10.5194/bg-21-3215-2024>
12. N. M. Travis, C. L. Kelly, & K. L. Casciotti. Testing the influence of light on nitrite cycling rates in eastern tropical North Pacific (2024). *Biogeosciences*, 21(8). <https://doi.org/10.5194/bg-21-1985-2024>
11. K. McMonigal, N. Evans, D. Jones, J. Brett, R. C. James, M. C. Arroyo, A. Y. Gong, E. C. Miller, C. L. Kelly, J. Middleton, C. Spear, W. Holmes, & D. Lane (2023). Navigating Gender at Sea. *AGU*

Advances, 4(4). <https://doi.org/10.1029/2023AV000927>

10. **C.L. Kelly**, C. Manning, C. Frey, J. Kaiser, N. Gluschkoff, & K.L. Casciotti (2023). pyisotopomer: A Python package for obtaining nitrous oxide isotopocules from isotope ratio mass spectrometry. *Rapid Communications in Mass Spectrometry*, 37(11). <https://doi.org/10.1002/rcm.9513>
9. N.M. Travis, **C.L. Kelly**, M.R. Mulholland, & K.L. Casciotti (2023). Nitrite Cycling in the Primary Nitrite Maxima of the Eastern Tropical North Pacific. *Biogeosciences*, 20(2). <https://doi.org/10.5194/bg-20-325-2023>
8. C. Frey, X. Sun, L. Szemlerski, K. Casciotti, E. García Robledo, A. Jayakumar, **C.L. Kelly**, M. Lehmann, & B.B. Ward (2023). Nitrous oxide production kinetics from ammonia oxidation in the Eastern Tropical North Pacific. *Limnology and Oceanography*, 68(2). <https://doi.org/10.1002/lno.12283>
7. P.J. Monreal*, **C.L. Kelly**, N.M. Travis, & K.L. Casciotti (2022). Identifying the sources and drivers of nitrous oxide accumulation in the eddy-influenced Eastern Tropical North Pacific oxygen-deficient zone. *Global Biogeochemical Cycles*, 36(6). <https://doi.org/10.1029/2022GB007310>
6. T. Cinay, D. Dumit, R.J. Woosley, E. Boles, J.V. Kwiecinski, S. Mullen, T.J. Tamasi, M.J. Wolf, **C.L. Kelly**, N.M. Travis, K.L. Casciotti, & A.R. Babbitt (2022). Coincident biogenic nitrite and pH maxima arise in the upper anoxic layer in the Eastern Tropical North Pacific. *Global Biogeochemical Cycles*, 36(12). <https://doi.org/10.1029/2022GB007470>
5. **C.L. Kelly**, N.M. Travis, P.A. Baya, & K.L. Casciotti (2021). Quantifying Nitrous Oxide Cycling Regimes in the Eastern Tropical North Pacific Ocean With Isotopomer Analysis. *Global Biogeochemical Cycles*, 35(2). <https://doi.org/10.1029/2020GB006637>
4. X. Sun, A. Jayakumar, J.C. Tracey, E. Wallace, **C.L. Kelly**, K.L. Casciotti, & B.B. Ward (2020). Microbial N₂O consumption in and above marine N₂O production hotspots. *The ISME Journal*, 15(5). <https://doi.org/10.1038/s41396-020-00861-2>
3. L.R. Damgaard, **C.L. Kelly**, K.L. Casciotti, B.B. Ward, & N.P. Revsbech (2020). Amperometric sensor for nanomolar nitrous oxide analysis. *Analytica chimica acta*, 1101. <https://doi.org/10.1016/j.aca.2019.12.019>
2. E. Haque, B.J. Mailloux, D. De Wolff, S. Gilioli, **C.L. Kelly**, E. Ahmed, C. Small, K.M. Ahmed, A. Van Geen, & B. Bostick (2018). Quantitative drinking water arsenic concentrations in field environments using mobile phone photometry of field kits. *Science of the Total Environment*, 618. <https://doi.org/10.1016/j.scitotenv.2016.12.123>
1. **C.L. Kelly** (2017). Dancing up the glass escalator: Institutional advantages for men in ballet choreography. *Columbia Undergraduate Research Journal*, 2(1). <https://doi.org/10.52214/curj.v2i1.4113>

Presentations

26. **C. L. Kelly**, "Capturing marine nitrogen cycling on new scales with machine learning," Environmental Geology and Geochemistry Seminar, Princeton University, September 2026, Princeton, NJ. *Invited talk*.
25. **C. L. Kelly**, "Low-pressure storms drive nitrous oxide emissions from the Southern Ocean," SOFLUX webinar, January 2026, online. *Invited talk*.
24. **C. L. Kelly**, "Biogeochemical and Physical Drivers of Air-Sea Gas Exchange: a Nitrous Oxide Case Study," University of Gothenburg Department of Marine Sciences, October 2025, Gothenburg, Sweden. *Invited talk*.
23. **C. L. Kelly**, "Insights into Southern Ocean Air-Sea Fluxes from Machine Learning and Autonomous Platforms," Marine Chemistry and Geochemistry Department Seminar, Woods Hole Oceanographic Institution, September 2025, Woods Hole, MA. *Invited talk*.
22. D.P. Nicholson, **C. L. Kelly**, & E. Park, "Large Scale Patterns of Biogeochemical Flux Inferred from the Biogeochemical Argo Array," 2025 Chemical Oceanography Gordon Research Conference, July 2025, Hooksett, NH. *Talk*.

21. **C. L. Kelly**, B.X. Chang, J.L. Bullister, M.J. Warner, N. Yoshida, S. Toyoda, A. Bourbonnais, M. de la Paz, D.H. Min, A. Makabe, S. Kawagucci, A. Barna, A. Macdonald, & D.P. Nicholson, "A comprehensive global marine nitrous oxide dataset from the Global Ocean Ship-Based Hydrographic Investigations Program," 2025 Chemical Oceanography Gordon Research Conference, July 2025, Hooksett, NH. *Poster*.
20. **C. L. Kelly**, B.X. Chang, A. Emmanuelli, E. Park & D.P. Nicholson, "Low-pressure storms drive N₂O outgassing from the Southern Ocean," 2025 Xiamen Symposium on Marine Environmental Sciences, January 2025, Xiamen, China. *Talk*.
19. **C.L. Kelly**, N.M. Travis, P.A. Baya, C. Frey, X. Sun, B. B. Ward, & K. L. Casciotti, "Isotopomer labeling and oxygen dependence of hybrid nitrous oxide production," 2025 Xiamen Symposium on Marine Environmental Sciences, January 2025, Xiamen, China. *Invited Talk*.
18. **C. L. Kelly**, "Low-pressure storms drive N₂O outgassing from the Southern Ocean," Applied Physics Lab Seminar Series, University of Washington, December 2024, Seattle, WA. *Invited talk*.
17. **C. L. Kelly**, "Low-pressure storms drive N₂O outgassing from the Southern Ocean," Ocean Carbon and Biogeochemistry (OCB) Seminar Series, December 2024, Virtual. *Invited talk*.
16. **C. L. Kelly**, B.X. Chang, A. Emmanuelli, E. Park & D.P. Nicholson, "Using machine learning and biogeochemical Argo floats to understand the ocean's role in nitrous oxide cycling," Ocean Carbon & Biogeochemistry Summer Workshop, June 2024, Woods Hole, MA. *Lightning talk*.
15. **C. L. Kelly**, B.X. Chang, J.L. Bullister, M.J. Warner, N. Yoshida, S. Toyoda, A. Bourbonnais, M. de la Paz, D.H. Min, A. Makabe, S. Kawagucci, A. Barna, A. Macdonald, & D.P. Nicholson, "A comprehensive global marine nitrous oxide dataset from the Global Ocean Ship-Based Hydrographic Investigations Program," 2024 Ocean Sciences Meeting, February 2024, New Orleans, LA. *Poster*.
14. **C. L. Kelly**, "Quantifying nitrous oxide cycling in an oxygen deficient zone: insights from isotopomers," Marine Chemistry and Geochemistry Department Seminar, Woods Hole Oceanographic Institution, September 2023, Woods Hole, MA. *Invited talk*.
13. **C. L. Kelly**, D. Bianchi, D. McCoy, S. Yang, B.X. Chang, & K.L. Casciotti, "Quantifying nitrous oxide production in the eastern tropical North Pacific hotspot by constraining process-based models with nitrous oxide isotopomers," 2023 Chemical Oceanography Gordon Research Conference, July 2023, Hooksett, NH. *Poster*.
12. **C. L. Kelly**, D. Bianchi, D. McCoy, S. Yang, & K.L. Casciotti, "Implementing isotopomers in process-based models reveals nitrous oxide sources in oxygen minimum zones," 2023 Association for the Sciences of Limnology and Oceanography Meeting, June 2023, Palma de Mallorca, Spain. *Invited talk*.
11. **C. L. Kelly**, "Quantifying Pacific Ocean nitrous oxide cycling using intramolecular isotope measurements and modeling," Chemistry Lunch Seminar, University of Washington, January 2023, Seattle, WA. *Invited talk*.
10. **C. L. Kelly**, "Quantifying Pacific Ocean nitrous oxide cycling using intramolecular isotope measurements and modeling," Dissertations Symposium in Chemical Oceanography (DISCO) XXVIII, October 2022, Kailua-Kona, HI. *Talk*.
9. **C. L. Kelly**, N. M. Travis, P. A. Baya, C. Frey, X. Sun, B. B. Ward, & K. L. Casciotti, "Identifying a potentially variable site preference for hybrid nitrous oxide production via isotopomer labeling experiments," 10th International Symposium on Isotopomers (ISI) & 12th Isotopes Conference, May 2022, Dübendorf, Switzerland. *Talk*.
8. **C. L. Kelly**, "Quantifying Pacific Ocean nitrous oxide cycling using intramolecular isotope measurements and modeling," Paleo/Environmental Seminar, University of Southern California, March 2022, Virtual. *Invited talk*.
7. **C. L. Kelly**, N. M. Travis, P. A. Baya, C. Frey, X. Sun, B. B. Ward, & K. L. Casciotti, "Isotopomer Labeling in Hybrid Nitrous Oxide Production," 2022 Ocean Sciences Meeting, February 2022, Virtual. *Talk*.

6. **C. L. Kelly**, C. Manning, C. Frey, N. Gluschankoff, & K.L. Casciotti, "pyisotopomer: A software package for N_2O isotopocule data corrections," 7th International Conference on Nitrification and Related Processes (ICoN7), July 2021, Virtual. *Poster*.
5. **C. L. Kelly**, N. M. Travis, P. A. Baya, C. Frey, X. Sun, B. B. Ward, & K. L. Casciotti, "Determining rates of hybrid archaeal N_2O production in the eastern tropical North Pacific Ocean with intramolecular isotope measurements," 2020 Ocean Sciences Meeting, February 2020, San Diego, CA. *Talk*.
4. **C. L. Kelly**, N. M. Travis, P. A. Baya, & K. L. Casciotti, "Quantifying nitrous oxide cycling regimes in the eastern tropical North Pacific Ocean with isotopomer analysis," What can we learn from N_2O isotope data?, October 2019, Dübendorf, Switzerland. *Talk*.
3. **C. L. Kelly**, N. M. Travis, P. A. Baya, & K. L. Casciotti, "Quantifying nitrous oxide cycling regimes in the eastern tropical North Pacific Ocean with isotopomer analysis," 2019 Gordon Research Conference on Chemical Oceanography, July 2019, Holderness, NH. *Invited talk*.
2. **C. L. Kelly**, B. X. Chang, C. Buchwald, M. S. Forbes, A. R. Babbin, & K. L. Casciotti, "Using Isotopomer Analysis to Determine Drivers of Nitrous Oxide Cycling in the Eastern Tropical North Pacific Ocean," 2018 Ocean Sciences Meeting, February 2018, Portland, OR. *Talk*.
1. **C. L. Kelly** & D. C. McCorkle, "Drivers of Seasonal and Interannual Variability in Waquoit Bay Carbonate Chemistry," 2017 ASLO Aquatic Sciences Meeting, February 2017, Honolulu, HI. *Poster*.

Proposals, Grants, and Fellowships

U.S. GO-SHIP Postdoctoral Fellowship (\$348,468).	2024-2025
WHOI Postdoc Scholar Award (\$106,000).	2023-2024
NSF Ocean Sciences Postdoctoral Research Fellowship (OCE-PRF) (<i>declined</i>).	2023
Ocean Sciences Meeting registration grant (\$355).	2022
Stanford Creative Expression Course Development Support (\$2,200).	2019
Stanford Student Projects for Intellectual Community Enhancement Grant (\$2,200).	2018
Stanford Woods Institute for the Environment Mel Lane Student Grant (\$1,500).	2018
Stanford Earth McGee-Levorsen Research Grant (\$3,000).	2018
National Science Foundation Graduate Research Fellowship Program (\$138,000)	2018
Stanford Enhancing Diversity in Graduate Education (EDGE) fellowship (\$11,800).	2017

Teaching and Mentoring

WHOI Machine Learning Seminar <i>Woods Hole Oceanographic Institution</i> Facilitated seminar series to share knowledge, best practices, and use cases of machine learning in oceanography.	Spring 2025-present
Undergraduate Student Mentor <i>Woods Hole Oceanographic Institution Guest Student Program</i> Guided an undergraduate student analyzing, interpreting, and applying machine learning models to autonomous glider data along the California Coast.	Summer 2025
Lab Hackathon <i>Woods Hole Oceanographic Institution</i> Led a two-day hackathon for the Biogeochemical Ocean Observing and Modeling Lab (BOOMLAB) at the Woods Hole Oceanographic Institution. Taught lab members how to run Python and Jupyter Notebooks on the computer cluster and developed code to predict nitrous oxide concentrations from Biogeochemical Argo float data.	May 2024

Undergraduate Student Mentor

Summer 2020-Spring 2022

*Stanford Earth Summer Undergraduate Research Program**Certificate of Achievement in Mentoring, Stanford School of Earth, Energy, and Environmental Sciences*

Guided an undergraduate student developing models of nitrous oxide cycling in an oxygen minimum zone, culminating in the undergraduate's first peer-reviewed, first author paper.

TA: Marine Chemistry

Spring 2020

Stanford University

Led weekly office hours via Zoom, graded, and taught chemical problem-solving skills.

TA: Atmosphere, Ocean, and Climate Dynamics: The Ocean Circulation

Winter 2020

Centennial TA Award, Stanford School of Earth, Energy, and Environmental Sciences

Led weekly in-person office hours, graded, helped students troubleshoot Python code in problem sets, and reviewed basic math concepts (partial differentials, integrals, etc.).

Academic Service**Co-Chair** WHOI Postdoc Association

2025-present

Co-Chair, Postdoc representative PRIDE@WHOI Employee Resource Group

2024-present

Lead WHOI nitrogen paper group

2023-2026

Chair-elect 2023 Gordon Research Seminar in Chemical Oceanography

2023

Special Service Award for Diversity, Equity, and Inclusion

2021

Stanford School of Earth, Energy, and Environmental Sciences.

Community Impact Award Stanford Alumni Association.

2020

Co-chair Stanford Earth Graduate Student Activities Committee (GSAC)

2019-2020

Department representative Stanford Earth Graduate Student Activities Committee (GSAC)

2018-2019

ODF Supertechnician GEOTRACES Pacific Meridional Transect (PMT, GP15)

2018

Peer reviewer PNAS, ISME, Nature Communications, Communications Biology, Geophysical Research Letters, Geoscientific Model Development, Global Biogeochemical Cycles, Biogeosciences, Limnology & Oceanography, Marine Chemistry, Geochimica et Cosmochimica Acta, Marine Environmental Research, Environmental Science & Technology, Atmosphere-Ocean, Regional Studies, One Earth**Art-Science Outreach Projects*****Cyclonic Flux and Argo Floats***

2024

Sculpture. Collaborated with artist Betsy Ritz as part of the Synergy project to create sculptures illustrating my postdoc research nitrous oxide fluxes in Southern Ocean storms, exhibited at the University of Rhode Island School of Oceanography August-September 2024.

Small Bugs in the Big City (Science Magazine Dance Your PhD Finalist)

2022

Video. Choreography and direction by R. Gellman with W-S. Law and C. Culha. Danced by **C. L. Kelly**, C. Culha, K. Foster, & W-S. Law, with R. Gellman. Filming/editing by C. O'Keefe. Sound editing by W-S. Law. Produced by R. Gellman, **C. L. Kelly**, & C. Culha. Based on R. Gellman's PhD work, "Differential carbohydrate metabolism impacts colonization by Hadza Bacteroidetes isolates." youtu.be/ut0Q7ubHsxs***N₂O in Drag***

2021

Video. Choreography and direction by **C. L. Kelly**. Danced by R. Gellman, C. Culha, & K. Foster, with **C. L. Kelly**. Filming/editing by C. O'Keefe. Music by S.C. Dobbs. Produced by **C. L. Kelly**, R. Gellman, & C. Culha. Based on C.L. Kelly's publication, "Quantifying Nitrous Oxide Cycling Regimes in the Eastern Tropical North Pacific Ocean With Isotopomer Analysis." youtu.be/R4KE-1KrpII

Deceptive Crystals

2021

Video. Choreography and direction by C. Culha. Danced by C. Culha, R. Gellman, & **C. L. Kelly**. Filming/editing by C. O'Keefe. Music by B. Gune. Produced by C. Culha, **C. L. Kelly**, & R. Gellman. Based on C. Culha's publications, "Crystal fractionation by crystal-driven convection" & "Evolution of thermal crystal zonations and their heterogeneity in crystal populations during magma cooling." youtu.be/H2cnAjppogg

Ocean Trilogy

2019

In-person performance. Choreography and direction by F. Spector-Atkins. Danced by the members of the SpectorDance company. Produced by **C. L. Kelly** & C. Culha. Based on interviews with scientists at the Monterey Bay Aquarium Research Institute.

Technical Skills

Programming & Development: Python (pandas, numpy, scikit-learn, xarray, matplotlib, seaborn, PyArrow), MATLAB, Git/GitHub, Shell scripting.

Machine Learning: Random Forest ensembles, supervised learning, feature engineering, cross-validation, uncertainty quantification.

Data Engineering: Large dataset processing, Parquet optimization, automated pipelines, time series analysis, Monte Carlo simulations.

High-Performance Computing: SLURM job scheduling, parallel processing optimization, cluster resource management, algorithm scaling.